Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Mateusz Balcerzak	Email	mateusz.balcerzak@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer IFAM Dresden			
Address	Winterbergstraße 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	*Hydrogen recovery/purification system*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Metal Materials*	*Energy Materials Technologies*
Concept of Proposed Technology	*The concept of the project is to develop and demonstrate a scalable hydrogen recovery/purification system based on metal-hydride technology, achieving TRL 6-7. The system will selectively separate hydrogen from mixed gas streams in industrial exhausts, with adaptability to different industrial sectors. The recovered/purified hydrogen can be then reused again at the industrial facility. Using metal hydrides, the solution will enable high-selective hydrogen separation/purification, and—where required—temporary storage and thermochemical compression (without mechanical compressors). The system will be designed as a modular, flexible product, allowing customization for different process gases and operating conditions. Validation will be performed using representative industrial use cases, demonstrating readiness for commercial deployment across multiple industries.*			

Rationale for Joint R&D Project	*Hydrogen is a critical process gas in numerous industrial sectors, including green steel production, semiconductor manufacturing, ammonia production, and biogas processing. In many of these processes, hydrogen is not fully consumed and is currently vented or underutilized; therefore, its recovery, purification, and reuse represent a significant economic and environmental opportunity for Korean industries.* *Moreover, according to Korean governmental strategy, H_2 will be blended into the natural gas pipeline and transported through existing national pipeline network (source KOGAS). In this context the proposed technology could be used as a solution for selective extraction of hydrogen from blended gas streams to supply purified hydrogen to industries and applications that require pure hydrogen for its operation.* *Fraunhofer IFAM brings nearly 20 years of experience in hydrogen-related technologies, with unique know-how and existing intellectual property (1-3) in metal-hydride-based hydrogen separation, storage, and compression. A joint R&D project between Fraunhofer IFAM and a Korean SME will enable effective technology and knowledge transfer, accelerating the deployment of advanced hydrogen recovery solutions tailored to the Korean industrial market.*
Key Words	1. *Hydrogen separation and purification* 2. *Hydrogen recovery in industrial processes* 3. *Adjustable and scalable separation unit* 4. *Metal hydrides*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	*The project will define generic system requirements for hydrogen separation/purification from industrial exhaust gases, including recovery/purification efficiency, gas composition tolerance, and scalability. Metal-hydride materials will be selected and optimized for selective hydrogen absorption and long-term cycling, using model exhaust gas compositions (for example exhaust gas from semiconductor industry as a representative use case). Engineering work will develop modular hydride-based separation/purification and/or, storage, and/or thermochemical compression units, including thermal management (with a possibility to integrate the developed system with industrial waste heat source) and control systems.* *Pilot-scale prototypes will be manufactured and tested in representative, model industrial gas environments (for example with semiconductor process exhausts serving as a validation scenario). Comprehensive testing will demonstrate performance, safety, adaptability, and TRL-7 readiness.*			
R&D Target Specifications	*The project targets the development of a modular metal-hydride-based hydrogen recovery and purification system achieving hydrogen product purities of 99 to 99.99 percent, depending on exhaust gas composition and reuse requirements. The system will be designed to recover at least 85 percent of the hydrogen contained in industrial exhaust gases with inlet hydrogen concentrations in the range of approximately 1 to 99 vol. percent (concentration can vary significantly depending on the chosen application sector and therefore composition of an exhaust gas). Stable operation can be demonstrated for representative mixed gas streams containing hydrogen together with nitrogen, argon, or methane, under feed pressures of approximately 1 to 30 bar and operating temperatures between 0 and 120 degrees Celsius (however, other pressure and temperatures are also possible).* *The separation and purification system will be based on standardized modules containing up to 10 kg of active metal hydride material per module. Each module is targeted to deliver approximately 0.1 to 0.3 kg of purified hydrogen per hour, depending on cycle time and operating conditions, enabling pilot scale systems with hydrogen production capacities in the range of several kilograms per day and straightforward scalability to higher throughputs by modular replication. The system will deliver purified hydrogen at outlet pressures of at least 5 to 30 bar, with the option to integrate thermochemical compression to reach pressures of up to approximately 80 bar, reducing or eliminating the need for mechanical compression.* *A key target of the project is a significant reduction in operating energy consumption through the use of available industrial waste heat for metal hydride regeneration. The system will be designed so that the majority of the thermal energy required for hydrogen release can be supplied by waste heat streams typically available in industrial facilities, in the temperature range of approximately 60 to 120 degrees Celsius. Depending on site-specific conditions and integration level, it is targeted that 70 to 100 percent of the thermal energy demand of the hydrogen separation and purification*			

	unit can be covered by waste heat, leading to a corresponding reduction of electrical energy consumption by approximately 70 to 100 percent compared to electrically heated operation. Any remaining electrical demand will be limited to auxiliary components such as control systems, valves, and pumps.
R&D Plan	*Overall approach and timeline (36 months total)* *The project is structured into two sequential phases, progressing from market and concept definition to pilot-scale development and validation of a modular metal-hydride-based hydrogen recovery and/or purification system.* *Phase I Pre-study (6 months)* • *Analysis of the Korean and global market for hydrogen recovery and/or purification technologies, including target industrial sectors, potential customers, competing solutions, and commercialization pathways* • *Definition of user and system requirements based on representative industrial exhaust gas streams and hydrogen reuse scenarios* • *Development of a basic engineering concept for a modular metal-hydride-based hydrogen recovery and/or purification, and optional compression system, including preliminary process flow and integration concepts* • *Preliminary techno-economic assessment and initial business case, including system sizing, cost drivers, energy savings potential, and value proposition for industrial users* • *Screening and preliminary selection of suitable metal hydride materials based on hydrogen capacity, selectivity, operating temperature, cycling stability, and compatibility with expected gas compositions* • *Phase I milestone: consolidated system concept, preliminary material selection, initial business case, and decision gate for Phase II* *Phase II Main R&D (30 months)* • *Experimental studies on selected metal hydride materials to optimize hydrogen absorption and desorption behavior, cycling stability, and tolerance to relevant impurities* • *Detailed system engineering including module design, thermal management concepts, control strategy, safety concept, and interfaces to industrial exhaust and utility systems* • *Design and construction of modular pilot-scale hydrogen recovery and purification units, including integration of waste heat utilization* • *Assembly of a complete pilot-scale system with scalable modular architecture* • *Testing and validation under representative, model industrial gas conditions, focusing on hydrogen purity, recovery efficiency, energy consumption, operational stability, and safety* • *Performance evaluation against defined target specifications and identification of optimization potential* • *Final milestone: pilot-scale demonstration achieving TRL 7 in representative conditions and preparation of documentation and roadmap for TRL 8 commercialization*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will take the significant role in the scientific and technological development of the metal-hydride-based hydrogen recovery/purification system. The institute will be responsible for defining system requirements, target specifications, and overall technical architecture based on representative industrial exhaust gas scenarios. Core responsibilities will include the screening, selection, and optimization of metal hydride materials, including experimental studies on hydrogen absorption and desorption behavior, cycling stability, and impurity tolerance. Fraunhofer IFAM will develop and validate key process concepts such as selective hydrogen separation, waste-heat-driven desorption, and optionally thermochemical compression. In addition, the institute will support the system-level engineering concept, including thermal management, safety concepts, and performance modeling. Fraunhofer IFAM will provide laboratory infrastructure, testing facilities, scientific personnel, and its existing know-how and intellectual property in hydrogen and metal hydride technologies, and will support pilot-scale testing, data evaluation, and TRL assessment.*
Expected R&R of Korean SME Partner	*The Korean SME partner will focus on application-driven development, engineering implementation, and market-oriented validation of the technology. The SME will contribute to market analysis, user requirement definition, and business case development, with particular emphasis on Korean industrial needs and regulatory conditions. Key responsibilities will include detailed mechanical and system engineering, manufacturing of modular pilot-scale units, and integration of the system into representative industrial process environments or test facilities. The SME will co-work on testing and validation activities, including long-duration operation, performance assessment, and reliability evaluation under realistic operating conditions. Furthermore, the SME will contribute engineering resources, manufacturing capabilities, and project management support, and will play a central role in preparing the pathway toward industrial deployment and commercialization in the Korean market in close collaboration with Fraunhofer IFAM. Favorable, Korean SME may be also responsible for the scale-up and mass production of metal hydrides (if one of the SMEs business case is related with alloy development and production) establishing manufacturing process and cost optimization of the active material (ant therefore the system) for industrialization. This would increase the manufacturability of the separation/purification unit by providing a stable active material supply and therefore decreasing the overall risk of the project.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The joint R&D project is expected to deliver a validated, modular hydrogen recovery/purification technology based on metal hydrides, demonstrated at pilot scale under representative industrial exhaust gas conditions. The key technological outcome will be a scalable system capable of selectively separating and purifying hydrogen to high purity levels, while enabling significant reductions in operating energy consumption through the utilization of industrial waste heat. The project will establish clear system performance data, engineering documentation, and safety concepts supporting TRL 7 validation and providing a solid basis for subsequent TRL 8 deployment in industrial environments.* *From a commercialization perspective, the project will create a market-ready technology platform that can be adapted to multiple industrial sectors, including semiconductor manufacturing, green steel production, ammonia and chemical processing, biogas upgrading and extraction of hydrogen from natural gas-based blends. The modular system architecture will allow flexible sizing and integration, lowering entry barriers for industrial users and supporting gradual market adoption. The involvement of a Korean SME ensures early alignment with market requirements, local regulations, and manufacturing capabilities, significantly accelerating technology transfer, localization, and commercialization in the Korean market. The Korean SME is expected to benefit from multiple income streams, including sales of modular hydrogen recovery systems, engineering and customization services for specific industrial applications, long-term operation and maintenance contracts, and potential licensing or co-licensing of developed technologies for regional markets.* *The broader industrial and economic impact includes improved resource efficiency and reduced hydrogen losses in energy-intensive industries, contributing to lower operating costs and reduced greenhouse gas emissions associated with hydrogen production. By enabling hydrogen reuse directly at industrial sites, the technology supports national and international hydrogen strategies, decarbonization goals, and circular economy principles. Furthermore, the project will strengthen Korean–German collaboration in hydrogen technologies, enhance the technological competitiveness of the participating SME, and contribute to the establishment of advanced hydrogen recovery solutions as a key enabling technology for sustainable industrial transformation.*

6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization